Semester 2 Examinations 2012/ 2013

Exam Code(s)	3BA1, 3BME1, 3FM2, 4BS4, 4CS2, 1HDS1
Exam(s)	3rd Arts, 3rd & 4th Science, H. Dip. (Appl. Sc)
Module Code(s)	MP307, MP341, MP391
Module(s)	Modelling II
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor G. Saccomandi
Internal Examiner(s)	Professor M. Destrade Dr P.T. Piiroinen
Instructions:	For full marks answer question 1 in Section A **and** TWO questions in Section B correctly. All questions carry the same weight.
Duration	2hrs
No. of Pages	5
Department(s)	Applied Mathematics
Course Co-ordinator(s)	Dr P.T. Piiroinen

Requirements:

Calculator	Yes
Handout	
Statistical/ Log Tables	Yes
Cambridge Tables	
Graph Paper	
Log Graph Paper	
Other Materials	

Section A - Answer all questions (a)-(d). (Each question is worth **5 Marks**.)

1. (a) Consider a queue and let

$$p_k = \rho^k(1-\rho), \quad 0 < \rho < 1 \quad \text{(Geometric distribution)}$$

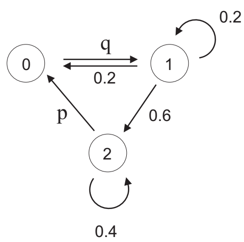
be the probability that k customers are queueing. If $\rho = \frac{1}{2}$,

(i) what is the mean queue length?

(ii) what is the variance and standard deviation of the queue length?

(b) (i) Describe a Markov process that is not ergodic? Explain your answer.

(ii) Consider the following Markov Process with transition probabilities shown.



Find the transition matrix P, the probabilities p and q and the equilibrium probabilities π_0, π_1 and π_2.

(c) Consider a model of two competing species N_1 and N_2 given by

$$\begin{aligned}\frac{dN_1}{dt} &= 100N_1 - 2N_1^2 - 2N_1N_2,\\ \frac{dN_2}{dt} &= 40N_2 - N_1N_2 - N_2^2.\end{aligned}$$

(i) Give an interpretation of the different terms in the model.

(ii) Find all equlibrium points for N_1 and N_2.

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= x_2 + b_1u,\\ \frac{dx_2}{dt} &= x_1 + x_2,\\ y &= x_1,\end{aligned}$$

where $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$ represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b_1 \in \mathbb{R}$ is the control parameter.

(i) Is this system observable? Explain your answer.

(ii) Is this system reachable? Explain your answer.

Section B - Answer any **two** questions.

2. A queue has a Poisson arrival pattern with parameter α_k and Poisson servicing pattern with parameter β_k where k is the queue size.

   (a) Assume that the maximum size of the queue is r.

   (i) Explain why $\alpha_r = \beta_0 = 0$.

   (ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

   $$\pi_k = \rho_k \rho_{k-1} \ldots \rho_1 \pi_0$$

   for $0 < k \leq r$ where $\rho_k = \alpha_{k-1}/\beta_k$.

   8 Marks

   (b) Assume the queue is infinite with

   $$\alpha_k = \alpha, \quad \beta_k = k\beta,$$

   where $\alpha > 0$ and $\beta > 0$.

   Show that the equilibrium probabilities are given by a Poisson distribution.

   6 Marks

   (c) Assume the queue corresponds to a telephone exchange with 4 operators. Each operator can deal with 1 customer enquiry per 2 minutes on average. If 30 calls per hour arrive on average what is the probability that the telephone exchange is saturated?

   6 Marks

3. Consider a species in which no individuals live beyond 3 days. Divide the population into four age groups labelled by $n = 0, 1, 2$ and 3. Assume that only the second ($n = 1$) and third ($n = 2$) groups can reproduce. For each group n, b_n denotes the birth rate and d_n the corresponding death rate.

   (a) Find a matrix $\mathbf{A}$ such that $\mathbf{P}(t+1) = \mathbf{A}\mathbf{P}(t)$ with

   $$\mathbf{P}(t) = \begin{pmatrix} P_0(t) \\ P_1(t) \\ P_2(t) \\ P_3(t) \end{pmatrix},$$

   where $P_n(t)$ denotes the population size of age group n at day t.

   4 Marks

   (b) Show that the eigenvalues of $\mathbf{A}$ satisfy the equation

   $$\lambda^4 - b_1(1-d_0)\lambda^2 - b_2(1-d_0)(1-d_1)\lambda = 0.$$

   4 Marks

   (c) If $d_0 = 0.1$, $d_1 = 0.2$, $d_2 = 0.4$, and $b_1 = 0.7, b_2 = 0.6$, show that there is one eigenvalue greater than 1. What is the significance of this?

   7 Marks

   (d) Verify your conclusion in (b) by considering this case over three days, where the initial population is $P_0 = P_1 = 100$ and $P_2 = P_3 = 50$.

   5 Marks

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_1 + c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0, p_1, p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the sum of the concentrations of the drug in compartments 1 and 2.

(a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.

4 Marks

(b) Investigate for what values of p_0, p_1 and p_2 this system is observable.

6 Marks

(c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 1$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -2$ and $\lambda_2 = -2$.

10 Marks